Case Study 01 Why Japan Recycles More Than Most Countries Lessons for India

Electronic Waste Management Assignment Report

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Executive Summary

Japan offers a useful example of assigning collection and recycling duties to identifiable actors. Separate arrangements cover specified large home appliances and small electrical items. India can adapt the convenient return routes and measured accountability while accounting for its own informal collection network.

Background

Discarded appliances contain metals, plastics, and components that need controlled handling. Japan’s home appliance framework places collection duties on retailers and recycling duties on manufacturers for specified products. Its small appliance law came into force in April 2013 to promote recovery from smaller devices [J1, J2].

Strategy and Initiative

The system gives consumers a defined return route, transfers appliances to approved processing, and measures recovery. Small devices can be aggregated through municipal arrangements and authorised operators. Separate product categories matter because a refrigerator needs different handling from a phone. India should test collection convenience, audit the destination, and publish product-wise results.

Impact and Results

Japan’s environment ministry reported about 14.95 million specified home appliances received by manufacturers in FY 2022. This is a count of units in that scheme, not a national e-waste recycling rate [J3]. The source supports collection scale but does not establish that every discarded device entered formal channels.

Indicator	Period	Result	Scope and source
Specified appliances received	FY 2022	About 14.95 million units	Japan environment ministry [J3]
Small appliance law effective	April 2013	National legal pathway	NIES [J2]

Indian national context	FY 2024–25	1,397,955.59 MT generated	PIB [I]

SWOT Analysis

Strengths	**Weaknesses**
Clear roles and established return route	Product coverage and fee-related participation can vary
Opportunities	**Threats**
Better collection design and product-wise tracking	Leakage through informal or improper disposal

Comparison with India

India’s 2022 rules use EPR obligations and registered recyclers, while Japan’s appliance routes emphasize identifiable collection and manufacturer processing. The published quantities use unlike units and product coverage, so they cannot be compared as national recycling rates [I, J1].

Recommendations

Create retailer return points for small devices; issue receipts and track each batch to a registered recycler; publish separate metrics for collection, reuse, recycling, and residual disposal; preserve jobs by enrolling existing collectors.

Conclusion

Japan’s main lesson is an understandable return route connected to responsible processing. India should strengthen that link and assess material outcomes rather than collection claims alone.

References

[J1] Japanese Law Translation, Small Waste Electrical and Electronic Equipment Act. https://www.japaneselawtranslation.go.jp/en/laws/view/3209/en

[J2] National Institute for Environmental Studies, Recycling Small Home Appliances and Metals in Japan. https://www-cycle.nies.go.jp/eng/column/page/202005_01.html

[J3] Japan Ministry of Environment, 2024 White Paper, FY 2022 appliance figure. https://www.env.go.jp/policy/hakusyo/r06/html/hj24020301.html

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 01 One Page Exam Revision Sheet

Why Japan Recycles More Than Most Countries Lessons for India

1. Two streams: specified large home appliances and small electrical products.

2. Stakeholders: consumers, retailers, manufacturers, municipalities, recyclers.

3. FY 2022: about 14.95 million specified appliances received by manufacturers.

4. Exam comparison: Japan has explicit return routes; India has EPR plus a large existing collector network.

5. Caution: units received are not a percentage of all e-waste generated.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 02 Switzerland E Waste Success Story

Executive Summary

Switzerland’s take-back model combines accessible collection, producer and importer finance, and specialist treatment. SENS data demonstrate substantial processing but cover the SENS system rather than all Swiss e-waste. India can borrow its convenience and reporting practices [S1].

Background

Consumers need a simple way to return unwanted equipment. SENS eRecycling is a non-profit Swiss organisation operating a take-back system, with advance recycling fees helping finance collection and recycling. Swico also operates a complementary system for electronic equipment [S1, S2].

Strategy and Initiative

A network of official collection points and retail returns channels equipment to contracted recyclers. Finance is arranged upstream, so the last holder faces fewer barriers at disposal. Reporting separates product groups and collection channels. Quality assurance and specialised treatment are essential for refrigerators, lamps, and devices with hazardous components.

Impact and Results

SENS reported 100,762 tonnes recycled in 2024, up from 92,874 tonnes in 2023, and 442 official SENS collection points in 2024. These figures are from SENS and must not be presented as the entire Swiss national total [S1].

Indicator	Period	Result	Scope and source
SENS recycled volume	2023	92,874 tonnes	SENS [S1]
SENS recycled volume	2024	100,762 tonnes	SENS [S1]
Official SENS collection points	2024	442	SENS [S1]

SWOT Analysis

Strengths	Weaknesses
Accessible collection and dedicated financing	Cost and system administration require oversight
Opportunities	**Threats**
Retail take-back and transparent channel data	Illegal diversion or poorly controlled downstream treatment

Comparison with India

India’s EPR framework similarly places responsibility on producers, but convenient return options and proof of physical recovery need wider local implementation. SENS tonnes cannot be divided by Indian generation to calculate a meaningful comparative rate [I, S1].

Recommendations

Map collection access by ward; require retailer and producer take-back publicity; publish annual volumes by product group; verify treatment records; budget for safe handling of hazardous parts.

Conclusion

The Swiss case shows that financing, easy returns, and processing standards must work together. India can adapt these elements through its EPR system and local partnerships.

References

[S1] SENS eRecycling, Facts and Figures 2024. https://www.erecycling.ch/ueber-sens/medien-und-publikationen/geschaeftsbericht/overview-2024/facts-figures.html

[S2] SENS eRecycling, organisation and take-back system. https://www.erecycling.ch/en/

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 02 One Page Exam Revision Sheet

Switzerland E Waste Success Story

1. Take-back = a clear path for consumers to return old equipment.
2. Advance recycling fee helps finance collection and treatment.
3. SENS 2024: 100,762 tonnes recycled; 442 official collection points.
4. India lesson: make the EPR return route easy to find and use.
5. Caution: SENS data are system-specific, not the total Swiss e-waste figure.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 03 India E Waste Management Rules 2022

Executive Summary

India’s E-Waste (Management) Rules, 2022 provide the main framework for covered electrical and electronic equipment. They came into force on 1 April 2023 and use registration, producer EPR obligations, and recycling certificates. The policy must be evaluated through physical waste flows and safe treatment, not registration counts alone [R1, R2].

Background

The rules replaced the 2016 framework for future operation and identify duties for manufacturers, producers, refurbishers, and recyclers. CPCB estimates generation from registered producer sales and average product life for notified categories; the estimate therefore has a defined scope [R1, R2].

Strategy and Initiative

Producers register and meet recycling obligations through the portal; registered recyclers process waste and generate certificates according to the rules. Refurbishing can extend useful life. Regulators should reconcile sales, stock, incoming scrap, recovered materials, certificates, and residue to test compliance.

Impact and Results

The government reported estimated generation of 1,254,286.55 MT in FY 2023–24 and 1,397,955.59 MT in FY 2024–25. The corresponding reported percentages collected, dismantled and recycled or disposed were 61.94% and 70.71%. This combined indicator is not a pure material recycling rate [R2].

Indicator	Period	Result	Scope and source
Estimated generation	FY 2023–24	1,254,286.55 MT	PIB [R2]
Combined managed share	FY 2023–24	61.94%	PIB [R2]
Estimated generation	FY 2024–25	1,397,955.59 MT	PIB [R2]
Combined managed share	FY 2024–25	70.71%	PIB [R2]

SWOT Analysis

Strengths	Weaknesses
Nationwide legal framework and digital records	Estimates and certificates need physical verification
Opportunities	**Threats**

Repair, collection expansion, and worker integration	Fraudulent certificates and unsafe processing

Comparison with India

Compared with Japan and Switzerland, India has a broad EPR regulatory structure but different product coverage, collection histories, and reporting units. Direct rankings would be misleading without consistent denominators [J1, S1, R2].

Recommendations

Publish clear definitions for each indicator; audit material balance at recyclers; connect informal collectors with registered processors; add consumer-facing return information and district-level access maps.

Conclusion

The rules are a necessary framework. Their practical success depends on convenient collection and verifiable, safe treatment of the equipment that enters the system.

References

[R1] MoEFCC, E-Waste (Management) Rules, 2022. https://moef.gov.in/uploads/2022/11/E-Waste-Management-Rules-2022.pdf

[R2] PIB, Parliament Question on E-Waste Processing Management, 24 July 2025. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 03 One Page Exam Revision Sheet

India E Waste Management Rules 2022

1. Effective date: 1 April 2023.

2. Actors: producer, manufacturer, refurbisher, recycler, regulator.

3. EPR: producer fulfils recycling obligation through the prescribed system.

4. FY 2024–25: 1,397,955.59 MT estimated generation; 70.71% combined managed indicator.

5. Exam caution: combined managed share is not a pure material recycling rate.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 04 India Extended Producer Responsibility Success or Paper Compliance

Executive Summary

EPR makes producers accountable for the post-use phase of covered products. Digital registration and certificates can scale oversight, but certificates alone cannot prove that each waste batch was physically and safely processed. The answer is to test records against material movements [R1, C1].

Background

Under India's 2022 rules, producers and recyclers participate in a portal-based system. Producers meet obligations through certificates generated by registered recyclers. Refurbishing and recycling have distinct roles. The framework responds to rising equipment sales and the need to recover materials without unsafe handling [R1, C1].

Strategy and Initiative

An effective audit reconciles producer obligations, recycler input weights, product types, process yields, residue, and destination documentation. Surprise inspections and sampling supplement portal returns. Citizen-facing take-back programmes supply genuine material to authorised facilities rather than just paperwork.

Impact and Results

Reported generation and the combined managed percentage increased between FY 2023–24 and FY 2024–25. These are national indicators, not evidence that every certificate is valid. A rigorous evaluation needs independent audits and incident reports [R2].

Indicator	Period	Result	Scope and source
Estimated generation	FY 2023–24	1,254,286.55 MT	PIB [R2]
Estimated generation	FY 2024–25	1,397,955.59 MT	PIB [R2]
Combined managed share	FY 2024–25	70.71%	PIB [R2]
Certificate-to-batch verification	Case evaluation	Independent audit required	Analytical indicator

SWOT Analysis

Strengths	Weaknesses
Clear producer obligation and digital trail	Portal records can diverge from actual processing

Opportunities	Threats
Third-party audit and traceable collection	Paper compliance, double counting, and unsafe diversion

Comparison with India

India’s EPR resembles the producer-financed logic of Switzerland, but the national portal alone does not create local consumer access. The Swiss example highlights the value of visible return points and treatment controls [S1, R1].

Recommendations

Require batch IDs from collection to final treatment; reconcile input and output quantities; disclose audit methodology and violations; reward repair where lawful; enable waste picker organisations to supply registered recyclers.

Conclusion

EPR can deliver real recovery when paperwork follows physical material. Compliance should be judged by verified treatment, collection access, and environmental and worker outcomes.

References

[R1] MoEFCC, E-Waste (Management) Rules, 2022. https://moef.gov.in/uploads/2022/11/E-Waste-Management-Rules-2022.pdf

[C1] CPCB, E-Waste EPR Frequently Asked Questions. https://cpcb.nic.in/uploads/Projects/E-Waste/FAQ_ewaste_23012024.pdf

[R2] PIB, E-Waste Processing Management. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 04 One Page Exam Revision Sheet

India Extended Producer Responsibility Success or Paper Compliance

1. EPR means the producer bears responsibility beyond sale.
2. Portal record → certificate → physical batch → recycler output → residue.
3. Risk: a certificate without matching real processing is paper compliance.
4. Control: material balance, inspections, third-party audits.
5. National figures show scale, not certificate validity.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 05 Bengaluru India E Waste Capital

Executive Summary

The “e-waste capital” label is a discussion prompt, not an established ranking in the supplied title list. Bengaluru’s concentration of electronics users and collection actors makes it a useful urban case. The key question is whether household and institutional waste reaches registered processing [B1, R2].

Background

Large urban technology markets generate discarded computers, peripherals, and household devices. Bengaluru’s municipal dry-waste collection infrastructure offers an entry point, though ordinary dry-waste totals cannot be treated as e-waste totals. Karnataka’s registered recycler processing is reported separately at state level [B1, R2].

Strategy and Initiative

Set up dedicated e-waste drop-off within accessible municipal and institutional collection routes. Train sorters to isolate devices and batteries, protect stored data, and record weights. Contract registered recyclers with documented handover and end treatment. Campus and office drives can complement household collection.

Impact and Results

Government data report 16,151.766 MT processed by registered recyclers in Karnataka in FY 2024–25, down from 20,910.603 MT in FY 2023–24. This is a state processing figure, not Bengaluru generation or city recycling performance [R2].

Indicator	Period	Result	Scope and source
Karnataka registered recycler processing	FY 2023–24	20,910.603 MT	PIB [R2]
Karnataka registered recycler processing	FY 2024–25	16,151.766 MT	PIB [R2]
Bengaluru-specific generation	This report	Not established	Data gap
Municipal dry-waste route	Current model	Collection centre pathway	BBMP [B1]

SWOT Analysis

Strengths	Weaknesses
Technology ecosystem and collection infrastructure	City-specific e-waste flow data are limited here

Opportunities	Threats
Campus, office and retailer take-back partnerships	Data leakage, unsafe informal processing, weak segregation

Comparison with India

India’s national EPR obligations apply in Bengaluru, but local success requires city-level access and proof of handover. Karnataka’s recycler volume is a state measure and must not be used to rank Bengaluru against Mumbai [R1, R2].

Recommendations

Publish a city e-waste mass balance; create accessible ward drop-offs; separate batteries and data-bearing devices; audit transfer to registered processors; include repair and reuse volumes.

Conclusion

Bengaluru’s opportunity is to connect high device turnover with convenient, accountable processing. A city ranking needs comparable city-level evidence.

References

[B1] BBMP Solid Waste Management, dry-waste collection pathway. https://site.bbmp.gov.in/departmentwebsites/swm/drywastekannada.html

[R2] PIB, state-wise e-waste processed. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 05 One Page Exam Revision Sheet

Bengaluru India E Waste Capital

1. Treat “capital” as a claim to test, not a measured fact.

2. State proxy: Karnataka registered recycler processing 16,151.766 MT in FY 2024–25.

3. City data needed: generated, collected, reused, recycled, residual.

4. Separate dry-waste collection statistics from e-waste statistics.

5. Key risk: lost traceability between collector and recycler.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 06 Mumbai E Waste Ecosystem

Executive Summary

Mumbai needs collection that works for housing societies, offices, repair markets, and smaller shops. Municipal dry-waste sorting centres and state-authorised recyclers can form a route, but a city-wide e-waste mass balance is needed to measure results [M1, M2].

Background

A dense city produces diverse discarded devices and depends on many intermediaries. BMC describes NGO-managed dry-waste sorting centres accepting recyclable material and e-waste in wards. MPCB publishes e-waste regulatory material and recycler information for Maharashtra [M1, M2].

Strategy and Initiative

Collect through ward centres, societies and institutional drives; protect data-bearing devices; identify items for repair or refurbishment; transfer the remainder with weighed receipts to registered processors. The city should map each handoff and distinguish e-waste from general dry waste in public reporting.

Impact and Results

Registered recyclers processed 52,597.803 MT in Maharashtra in FY 2024–25 versus 40,664.231 MT in FY 2023–24. Those are statewide figures, not Mumbai-only collections or a measure of BMC’s programme [R2].

Indicator	Period	Result	Scope and source
Maharashtra registered processing	FY 2023–24	40,664.231 MT	PIB [R2]
Maharashtra registered processing	FY 2024–25	52,597.803 MT	PIB [R2]
Mumbai-only e-waste generation	This report	Not established	Data gap
Ward sorting model	BMC description	NGO-managed centres	BMC [M1]

SWOT Analysis

Strengths	Weaknesses
Dense collection networks and institutional demand	Fragmented handoffs and limited city-specific metrics
Opportunities	**Threats**

Ward-level routing and formal repair networks	Unsafe stripping and leakage to untracked disposal

Comparison with India

Mumbai operates within India's EPR rules. Its practical bottleneck is the local collection and transfer chain, similar to other Indian cities; Switzerland offers a useful example of easy returns and published system quantities [R1, S1].

Recommendations

Make ward drop-off information public; contract collectors with receipts; track device categories and weights; support repair before material recovery; verify authorised downstream treatment and worker protections.

Conclusion

A strong Mumbai ecosystem must join existing collection and repair activity to safe processing. City-level results should be published separately from Maharashtra totals.

References

[M1] Brihanmumbai Municipal Corporation, solid waste initiatives. https://www.mcgm.gov.in/irj/portal/anonymous/qlcleanover?guest_user=english

[M2] Maharashtra Pollution Control Board, e-waste management. https://www.mpcb.gov.in/en/node/4297

[R2] PIB, state-wise e-waste processed. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 06 One Page Exam Revision Sheet

Mumbai E Waste Ecosystem

1. Actors: households, offices, ward centres, repairers, recyclers, BMC, MPCB.
2. Maharashtra registered processing: 52,597.803 MT in FY 2024–25.
3. State processing is not Mumbai collection.
4. Trace chain: source → centre → repair/refurbishment → authorised recycler.
5. Need Mumbai-specific mass balance and worker safety reporting.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 07 Kerala Clean Kerala Company

Executive Summary

Kerala's decentralised waste collection can link household sorting, local self-government, Haritha Karma Sena, and Clean Kerala Company. The broader inorganic-waste programme illustrates the network; e-waste volumes must be reported separately [K1, K2].

Background

Haritha Karma Sena workers collect and segregate non-biodegradable waste through local bodies. State and local sources identify Clean Kerala Company as a downstream partner. Some local collection calendars explicitly include e-waste, but ordinary inorganic waste data cannot be treated as e-waste [K1, K2].

Strategy and Initiative

Use scheduled household collection, material collection facilities, separate storage for electrical devices, and documented handover to authorised processors. Provide training, protective equipment, and payment clarity to collection workers. Repairable devices should be assessed before destructive recycling.

Impact and Results

A Kerala local government report says 50,190 tonnes of inorganic waste were handed to Clean Kerala Company over one year by 35,214 women in 4,438 Haritha Karma Sena units. These are broad waste-system figures, not e-waste-specific results. Government data also show 10,079.632 MT processed by registered e-waste recyclers in Kerala in FY 2024–25; this is statewide and not solely Clean Kerala Company [K1, R2].

Indicator	Period	Result	Scope and source
Inorganic waste handed to CKC	Reported one-year period	50,190 tonnes	LSGD [K1]
HKS workers / units	Same report	35,214 / 4,438	LSGD [K1]
Kerala registered e-waste processing	FY 2024–25	10,079.632 MT	PIB [R2]
CKC-specific e-waste volume	This report	Not established	Data gap

SWOT Analysis

Strengths	Weaknesses
Local workforce and decentralised collection	Mixed-waste totals obscure e-waste performance
Opportunities	**Threats**

Scheduled device collection and formal handover	Contamination, unsafe storage, uneven local capacity

Comparison with India

Kerala's strength within India is the local collection workforce. National EPR obligations still require authorised downstream processing; the state network can supply the collection layer that an EPR certificate cannot create by itself [R1, K1].

Recommendations

Add a separate e-waste register to every local collection facility; publish CKC-specific device volumes and destinations; ensure fire-safe storage; pay collectors fairly and train them on batteries and data devices.

Conclusion

Kerala's model shows how local workers can make collection reachable. Its e-waste impact should be evaluated with dedicated e-waste metrics rather than broad inorganic waste totals.

References

[K1] Kerala Local Self Government Department, Haritha Karma Sena report. https://lsgkerala.gov.in/index.php/en/kudumbashree/news/kudumbashree-haritha-karma-sena-kerala-model-cleanliness

[K2] Ongallur Grama Panchayat, Haritha Karma Sena and MCF. https://ongallurpanchayat.lsgkerala.gov.in/en/page/haritha-karma-sena

[R2] PIB, state-wise e-waste processed. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 07 One Page Exam Revision Sheet

Kerala Clean Kerala Company

1. Kerala chain: household → HKS → MCF → CKC/downstream partner.
2. 50,190 tonnes is all inorganic waste in the cited report, not e-waste.
3. Kerala registered e-waste processing: 10,079.632 MT in FY 2024–25.
4. Measure e-waste separately by device type and final destination.
5. Include worker training, payment, and safe storage.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

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Case Study 08 Pune SWaCH Model

Executive Summary

SWaCH is a waste picker-owned cooperative working with Pune Municipal Corporation. Its model makes social inclusion part of collection. SWaCH reports e-waste collection authorisation in 2012 and use of bins and drives, with downstream channelisation to authorised partners [P1, P2].

Background

Waste pickers have long recovered value from discarded materials but often lack stable recognition and protection. The SWaCH partnership grew from a pilot involving 1,500 waste pickers and 125,000 households, according to SWaCH. Those numbers describe the early door-to-door waste collection pilot, not e-waste throughput [P1].

Strategy and Initiative

Residents and organisations can use e-waste bins and drives. Workers aggregate devices, and authorised partners handle recycling. The model works best when collection receipts, safe handling training, and transparent payments follow the same chain. SWaCH's broad waste services should not be confused with a measured e-waste-only output.

Impact and Results

SWaCH reports its formation in 2007 and e-waste collection authorisation in 2012. Its pilot involved 1,500 workers and 125,000 households. No comparable verified SWaCH e-waste tonnage is used in this report [P1]. Maharashtra recycler processing is available as statewide context only [R2].

Indicator	Period	Result	Scope and source
Pilot waste pickers	2005 pilot	1,500	SWaCH [P1]
Pilot households served	2005 pilot	125,000	SWaCH [P1]
Cooperative formed	2007	SWaCH established	SWaCH [P1]
E-waste collection authorisation	2012	Reported by SWaCH	SWaCH [P1]

SWOT Analysis

Strengths	Weaknesses
Worker ownership and established door-to-door access	E-waste-specific public output data need strengthening
Opportunities	**Threats**

Scale collection through societies and institutions	Unsafe handling or exclusion through poorly designed contracts

Comparison with India

India's EPR rules focus on producer and recycler duties. SWaCH shows how worker organisations can provide the citizen-facing collection layer, improving access and livelihoods while routing materials to registered recyclers [R1, P1].

Recommendations

Publish annual e-waste weights and destinations; pay workers for service and material value; provide safety and data-erasure protocols; use formal contracts and audit partner recyclers.

Conclusion

The Pune model links environmental management with livelihoods. Its strongest transferable lesson is to integrate workers into traceable, safe e-waste collection.

References

[P1] SWaCH, SWaCH–PMC Partnership. https://swachcoop.com/about/swach-pmc-partnership/

[P2] SWaCH, Zero Waste e-waste service. https://swachcoop.com/zero-waste/

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 08 One Page Exam Revision Sheet

Pune SWaCH Model

1. SWaCH = waste picker-owned cooperative in Pune.
2. Early pilot: 1,500 workers and 125,000 households; not e-waste totals.
3. Formed in 2007; e-waste collection authorisation reported for 2012.
4. Model: inclusive collection → documented handover → authorised recycler.
5. Exam point: EPR needs a reachable local collection network.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 09 Dharavi Recycling Ecosystem

Executive Summary

Dharavi is a useful case for analysing repair, reuse, small traders, and informal material recovery. The supplied topic list does not contain a Dharavi-specific e-waste survey or verified throughput. This section therefore evaluates an ecosystem and specifies the evidence needed before making quantitative claims.

Background

Informal and small-scale enterprises often collect, test, repair, resell, and dismantle devices. Reuse can extend product life, while unprotected handling and uncontrolled material extraction can harm workers and surroundings. An e-waste analysis should distinguish electrical devices from the broader recycling economy and avoid repeating unverified turnover or tonnage claims.

Strategy and Initiative

Map collectors, repair shops, brokers, and final destinations. Offer shared testing, protective equipment, safe storage, data-wiping support, and access to registered recyclers. Preserve viable reuse and livelihoods while preventing burning, uncontrolled acid use, or disposal of hazardous residuals. A partnership should have documented weights and fair prices.

Impact and Results

No defensible Dharavi-only e-waste quantity or recycling rate was established from the supplied titles or sources used here. Maharashtra registered recycler volumes give state context but do not measure Dharavi. The table makes that distinction explicit [R2].

Indicator	Period	Result	Scope and source
Dharavi-specific e-waste generation	This report	Not established	Primary survey needed
Dharavi-specific recovered volume	This report	Not established	Primary survey needed
Maharashtra registered processing	FY 2024–25	52,597.803 MT	State context, PIB [R2]
Worker safety incidents	This report	Not established	Health and safety study needed

SWOT Analysis

Strengths	Weaknesses
Repair skills and dense reuse networks	Limited verified flow and safety data

Opportunities	Threats
Formal partnerships without loss of livelihoods	Toxic exposure, displacement, and untracked residue

Comparison with India

India’s rules define registered formal processing, while Dharavi illustrates the importance of existing market networks before recycling. Policy should not equate all informal activity with unsafe recycling; repair and collection can be connected to compliant downstream facilities [R1].

Recommendations

Conduct a device-specific baseline survey; co-design collection and processing links with workers; provide PPE and health monitoring; document material prices and final destinations; measure reuse separately from recycling.

Conclusion

The lesson is to formalise safety and traceability while retaining repair value and livelihoods. Claims of Dharavi’s e-waste performance require a dedicated survey.

References

[R1] MoEFCC, E-Waste (Management) Rules, 2022. https://moef.gov.in/uploads/2022/11/E-Waste-Management-Rules-2022.pdf

[R2] PIB, state-wise registered e-waste processing. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

[M2] Maharashtra Pollution Control Board, e-waste resources. https://www.mpcb.gov.in/en/node/4297

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 09 One Page Exam Revision Sheet

Dharavi Recycling Ecosystem

1. Dharavi case = repair, reuse, trade, collection and possible downstream recycling.
2. Do not invent Dharavi e-waste tonnage or recycling rate.
3. Survey: source, device type, weight, worker exposure, buyer and final destination.
4. Strength: livelihoods and repair; threat: unsafe extraction.
5. India lesson: integrate collectors and repairers with registered processors.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

Common Exam Mistake

Do not confuse a target with an achieved result, a state figure with a city result, general dry waste with e-waste, or a combined managed percentage with a pure recycling rate.

Case Study 10 Dell Circular Economy Strategy

Executive Summary

Dell approaches e-waste through product design, take-back, reuse, recovery, and recycled materials. Its public 2030 commitments show direction; a commitment is not an achieved result. Product examples demonstrate material choices, but they do not prove the environmental impact of all products [D1, D2].

Background

Manufacturers influence repairability, component replacement, material choices, and end-of-use collection. A circular strategy aims to keep products and materials useful for longer and to reduce the need for virgin inputs. Dell describes design for disassembly and recovery services [D1].

Strategy and Initiative

Design devices so components can be repaired or extracted; collect used equipment through take-back; refurbish where suitable; recycle remaining materials responsibly; use recovered content in new products. Measure output with independently defined indicators, including take-back weight, reuse, recycled input, and residual disposal.

Impact and Results

Dell states 2030 goals of reusing or recycling one metric ton for every metric ton sold, packaging made from recycled or renewable materials or reused packaging, and more than half of product content from recycled, renewable, or reduced-emissions materials. These are targets, not completed outcomes [D2]. Dell also reports a Latitude 9440 example with 75% recycled content in its low-emissions aluminium chassis and 90% post-consumer recycled plastic in the battery housing [D3].

Indicator	Period	Result	Scope and source
Take-back/reuse/recycling goal	By 2030	1 tonne per tonne sold	Target, Dell [D2]
Packaging goal	By 2030	100% qualifying material or reuse	Target, Dell [D2]
Latitude 9440 chassis	Product example	75% recycled content	Dell [D3]
Latitude 9440 battery housing	Product example	90% PCR plastic	Dell [D3]

SWOT Analysis

Strengths	Weaknesses
Design control and global take-back	Targets need transparent outcome

capability	verification
Opportunities	**Threats**
Repairable design and closed-loop material recovery	Complex supply chains and rebound from new device sales

Comparison with India

India’s EPR rules assign producer recycling obligations. Dell’s circular approach goes further upstream to product design and material selection, which Indian producers can adopt while complying with local registered recycling and take-back rules [R1, D1].

Recommendations

Publish annual achieved figures separately from 2030 goals; measure product life and repair rates; expand easy returns in India; use audited recycled-content claims; ensure responsible processing and data destruction.

Conclusion

The manufacturer can reduce future waste through better design and take-back. The decisive test is independently verifiable progress across product life, reuse, and material recovery.

References

[D1] Dell Technologies, Accelerating the Circular Economy. https://www.dell.com/en-in/dt/corporate/social-impact/advancing-sustainability/accelerating-the-circular-economy.htm

[D2] Dell Technologies, Circular Economy in Action, 2024. https://www.dell.com/en-us/blog/circular-economy-in-action-leading-the-fight-against-e-waste/

[D3] Dell Technologies, Driving Sustainable Impact, Latitude 9440 example. https://www.dell.com/en-us/blog/how-were-driving-sustainable-impact-for-business-and-society/

[I] PIB, national e-waste figures. https://www.pib.gov.in/PressReleasePage.aspx?PRID=2147876

Case Study 10 One Page Exam Revision Sheet

Dell Circular Economy Strategy

1. Circular sequence: design → use/repair → take-back → reuse → recycling → recycled input.

2. Dell 2030 one-for-one metric ton is a goal, not an outcome.

3. Example: Latitude 9440 chassis 75% recycled content; battery housing 90% PCR plastic.

4. Compare with India: EPR is a regulatory duty; design choices act earlier.

5. Exam caution: distinguish target, product example, and achieved company-wide result.

Quick Answer Structure

Define the e-waste problem and stakeholders. Explain collection and downstream treatment. Add the table figure with its exact geography and period. Discuss strengths, limits, India comparison, and two practical recommendations.

Terms to Remember

EPR: extended producer responsibility. Refurbishment: extending useful product life. Recycling: recovering material from discarded equipment. Traceability: records linking origin, transfer, processing, and final treatment.

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